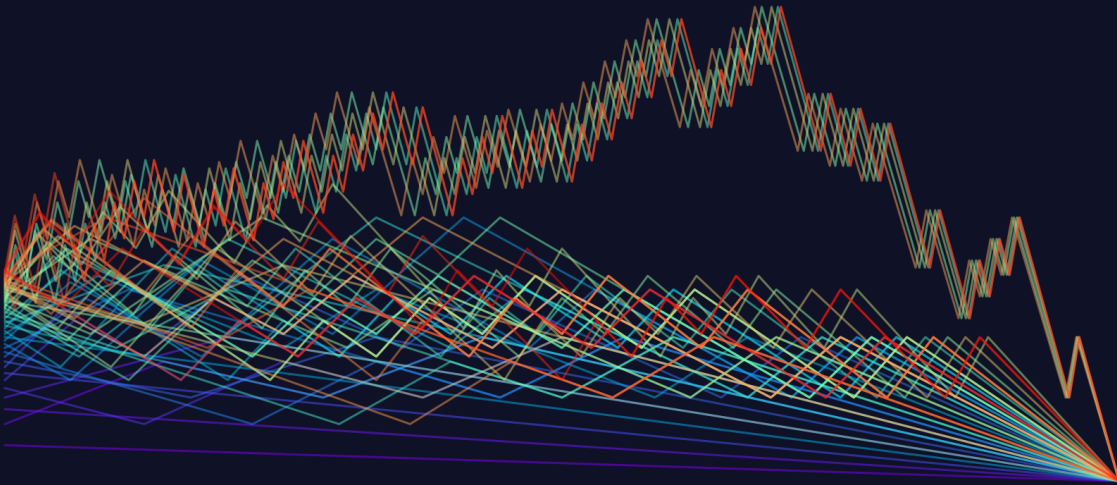


THE COLLATZ CONJECTURE

Simple to State. Unsolved for Nearly a Century.



Pick any positive whole number. If it's even, halve it. If it's odd, triple it and add one. Repeat forever — and, it seems, you always land on 1. No one has ever found an exception. No one has ever proven there isn't one.

What Is the Collatz Conjecture?

The Collatz Conjecture is one of the most famous unsolved problems in mathematics — and one of the easiest to explain to a child. It was proposed by German mathematician Lothar Collatz in 1937, and it concerns a simple process applied repeatedly to positive integers.

The Rule

Start with any positive integer n . Then repeat the following two steps:

If n is even	Divide it by two: $n \rightarrow n / 2$
If n is odd	Triple it and add one: $n \rightarrow 3n + 1$

Keep applying this rule to whatever number you get. The conjecture claims that no matter which positive integer you start with, you will eventually reach 1. Once a sequence hits 1, it falls into the repeating loop $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$ forever.

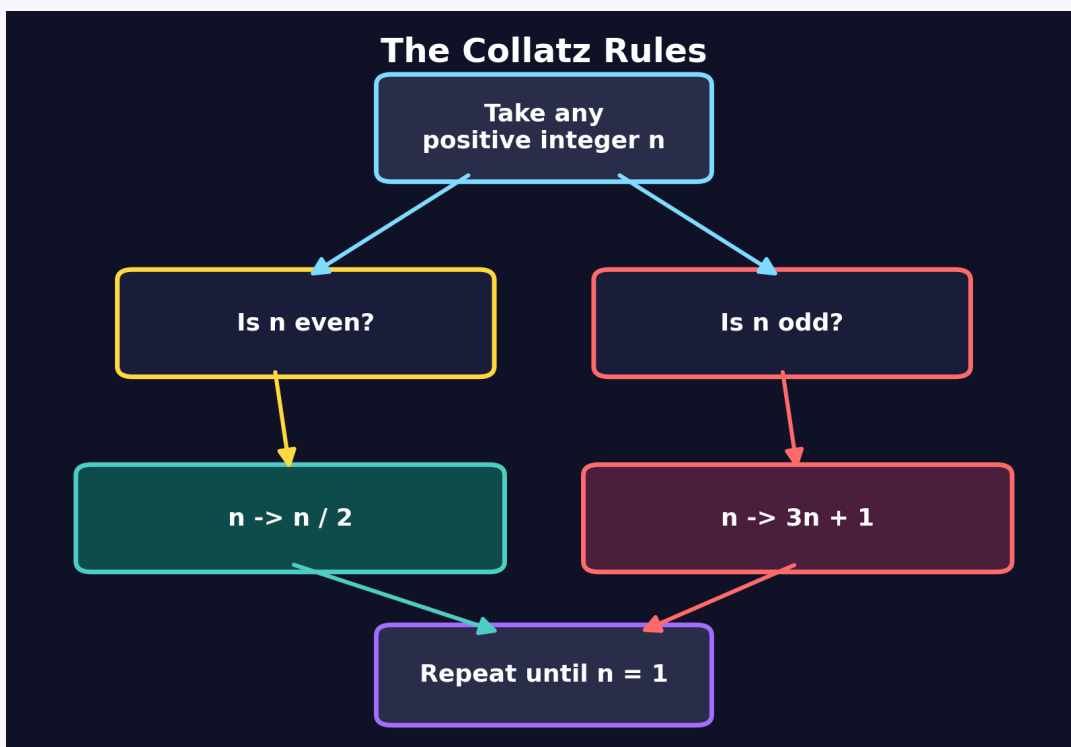


Figure 1: The two simple rules that define every Collatz sequence.

A Worked Example: Starting at 27

Small starting numbers can produce surprisingly long and wild journeys. The number 27, for instance, takes 111 steps and climbs as high as 9,232 before finally collapsing down to 1. Here is the beginning of its sequence:

27, 82, 41, 124, 62, 31, 94, 47, 142, 71, 214, 107, 322, 161, 484, 242, 121, ...

This up-and-down, seemingly chaotic bouncing before an eventual descent to 1 is why these sequences are often called **hailstone numbers** — like a hailstone in a storm cloud, bounced up and down repeatedly before finally falling to the ground.

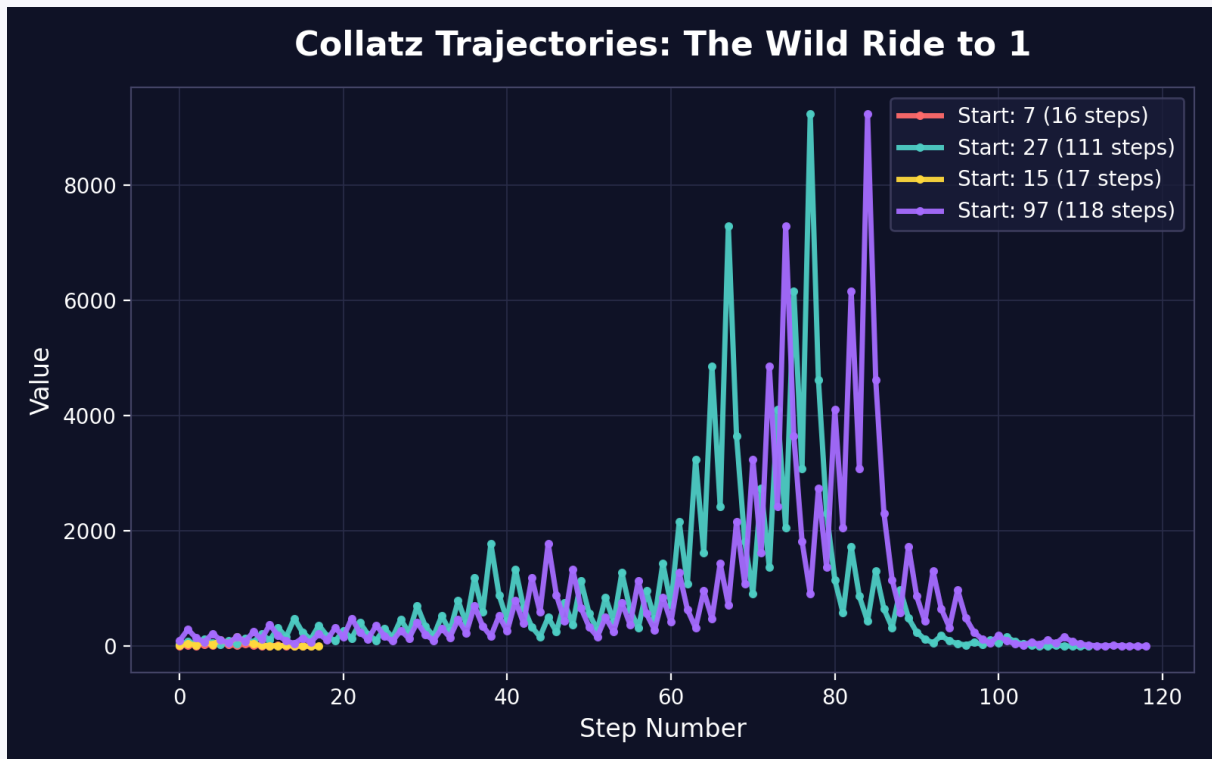


Figure 2: Four different starting numbers, four wildly different journeys — yet all of them plunge to 1 in the end.

Looking for Patterns

Mathematicians have verified the conjecture by computer for every starting number up to about 2^{68} (roughly 300 quintillion) — every single one eventually reaches 1. The chart below shows how many steps it takes for the numbers 1 through 60. There is no obvious formula that predicts the length just from the starting number.

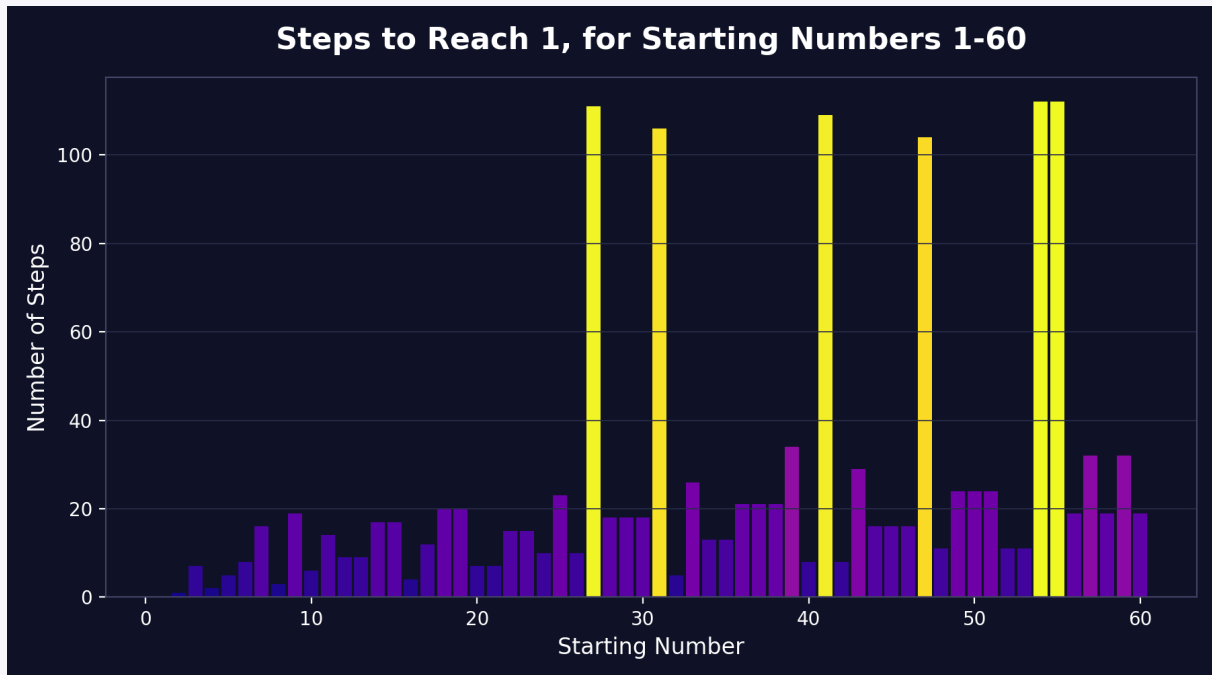


Figure 3: Steps to reach 1 for starting numbers 1-60. Nearby numbers can have wildly different stopping times.

Zooming out to the first 10,000 starting numbers reveals a strange, fractal-like texture: distinct streaks and bands emerge, hinting at deep structure, yet still no one has found a pattern that explains — let alone proves — why every sequence must terminate.

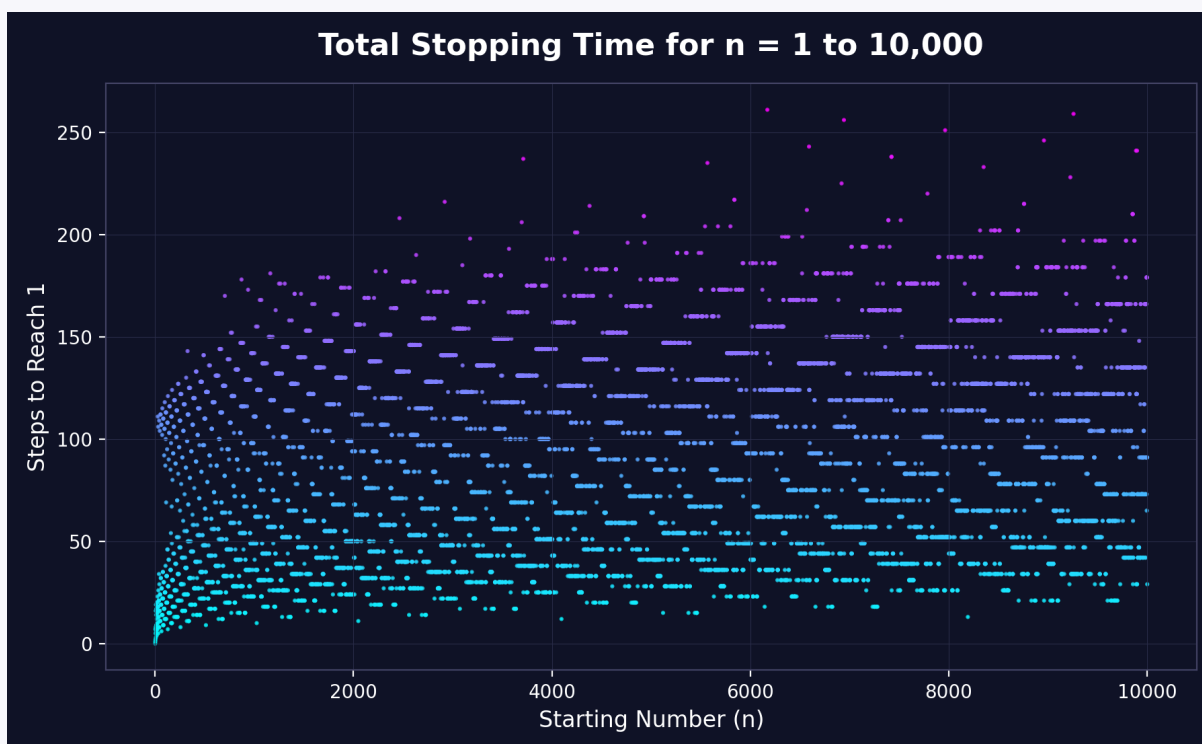


Figure 4: Total stopping time for the first 10,000 positive integers, colored by step count. The banded, fractal-like structure is characteristic of the problem.

Why Is This So Hard to Prove?

The Collatz Conjecture mixes two very different operations — halving (a smooth, predictable, base-2 operation) and tripling-plus-one (which depends on a number's digits in an unrelated way). This clash between multiplicative structure and additive structure makes the sequences behave almost randomly, even though they are completely deterministic. That blend of simplicity and chaos is part of what makes the problem so seductive — and so resistant to proof.

Mathematician Paul Erdos is widely quoted as saying that mathematics may not yet be ready for such problems, and Jeffrey Lagarias, who has studied the conjecture extensively, has described it as an exceptionally difficult problem, completely out of reach of present-day mathematics.

Key Facts

- Proposed by Lothar Collatz in 1937.
- Also known as the $3n + 1$ problem, the Ulam conjecture, and the Syracuse problem.
- Verified by computer for all starting values up to roughly 2.95×10^{20} (as of recent large-scale computations), with no counterexample ever found.
- In 2019, Fields Medalist Terence Tao proved that the conjecture is true for "almost all" positive integers in a precise statistical sense — a major partial result, though not a full proof.
- No one has proven the conjecture true, and no one has found a counterexample.

Why Mathematicians Care

Beyond its surface simplicity, the problem touches on deep questions in number theory and dynamical systems. It has become a benchmark for testing new mathematical tools, and it is a favorite example for illustrating how a problem a child can understand can still be beyond the reach of the world's best mathematicians.

Try it yourself: pick any number and follow the rule. However long it takes, it always seems to end the same way — at 1.